AVISHKARNA INTERNSHIP PROGRAM

Mini Project Submission Format

Student Name:	Bhanu venkat
College Name:	NRI INSTITUTE OF TECHNOLOGY
Branch & Year:	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING_3rd year
Roll Number:	23KP1A6151
Email ID:	bhanuvenkatsingamsetty@gmail.com
Mobile Number:	9885538072

Project Title:	RetailEdge Sales Report & Executive KPI Dashboard
Project Domain:	MACHINE LEARNING

Project Objective

The objective of this project is to automate the ingestion, cleaning, and reporting of retail transaction data. The system standardizes raw logs, performs dynamic catalog lookups, and automatically builds structured Excel sheets featuring charts, custom formulas, and pivot analyses. It also publishes an interactive React web dashboard to present key executive performance summaries to stakeholders.

Components / Software Used

Programming & Scripting: Python 3.11, JavaScript (React 19, Vite, Recharts, Lucide)
Libraries/APIs: win32com.client (pywin32) for Microsoft Excel automation, json, os, random
Storage: Excel (.xlsx) spreadsheets, dashboard_data.json data bridge
Tools: Git, GitHub, VS Code editor

Working Principle

The project has a pipeline of four automated stages:
1. *Data Generation:* A script generates synthetic transaction logs with messy string values.
2. *Excel Processing:* Python calls the local Microsoft Excel COM interface to clean regions and run lookups.
3. *Report Rendering:* Excel compiles three pivot sheets, creates custom visual trend charts, and outputs a flat text report.
4. *Dashboard visualization:* A local Python web server displays a React dashboard loading the live data.

Results

• Professionally formatted **RetailEdge_Sales_Report.xlsx** featuring dynamic pivot tables, formatting rules, and visual charts.
• Structured summary text file **RetailEdge_KPI_Report.txt** identifying critical sales, averages, and territory peaks.
• Live-updating interactive web dashboard showing month-by-month and division breakdowns.

Project Submission

Source code and pre-compiled dashboard assets are published at the GitHub link below.

Source Code Link

https://github.com/Bhanu13060/Retail-sales-data

Demo Video

____ (Optional)

Challenges Faced

• Rebuilding a broken virtual environment that had absolute paths hardcoded for a foreign machine.
• Transitioning absolute outputs in the Excel scripting to compute paths relative to the current file location.
• Providing access to the React dashboard on machines missing Node.js by scripting a local Python HTTP fallback server.

Future Scope

• Upgrading data pipelines to interface directly with active SQL databases instead of generated static listings.
• Implementing email triggers in the python script to automatically distribute reports to managers.
• Restructuring dashboard routes to include role-based login authorization.

Declaration

I declare that this project is my own work.

Student Signature: Bhanu venkat **Date: July 28, 2026**